**SIMATS ENGINEERING**
**ASSESSMENT TOOL 2:**

# Case Study

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

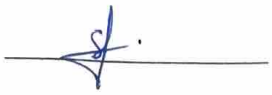
Dave's Taxonomy: Precision (Level 3)
Question Count: 3

**Total Marks: 30**

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**Code of Conduct**

I S.SHAMAH ALI FATHIMA (192573002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

## Assessment Tasks

# Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

   Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

**Case Study Rubric**

| **Criteria** | **Weightage** | **Level 4 - Exemplary** | **Level 3 - Proficient** | **Level 2 - Developing** | **Level 1 - Beginning** |
|---|---|---|---|---|---|
| Understanding of Case | 25% | Clearly identifies all technical and business requirements | Identifies most requirements | Basic understanding only | Poor understanding of case |
| Application of Concepts | 25% | Applies suitable cloud concepts and tools effectively | Good application with minor issues | Partial application | Weak or incorrect application |
| Analysis | 20% | Strong analysis with clear trade-off reasoning | Reasonable analysis | Limited analysis | Minimal analysis |
| Solution Design | 20% | Solution is feasible, practical, and well justified | Reasonable solution with minor gaps | Basic solution | Unclear solution |
| Clarity | 10% | Well-structured and easy to follow | Mostly clear | Somewhat unclear | Poorly organized |

SE STUDY 1 :

A smart retail inventory management system uses cloud computing to manage product stock, sales transactions, and inventory across multiple stores.

Cloud - Based solution :

i) client layer :

Pos systems and dashboards allow employees to record sales and check stock

ii) web API Layer :

REST APIs, transfer product and transaction data between stores and cloud services.

iii) Cloud application layer :

Processes transactions and updates stock application

iv) Cloud storage :

Stores product details, Inventory records and transaction logs.

Working Process:

POS System → REST API → cloud Application → cloud Database → Analytics and reports.

* when a product is sold, the POS System sends transaction details to cloud.

* The application updates the stock database and generates alerts when inventory becomes low.

2) CASE STUDY 2: CLOUD BASED DOCUMENT COLLABORATION SYSTEM.

INTRODUCTION:

A cloud-based Document collaboration system allow multiple users to create, edit and share documents through webbrowsers.

Cloud Based Solution:

i) Client layer:

Browsers and mobile devices allow users to edit and share documents.

ii) API Layer:

Restart APIs support document creation, retrieval and update.

iii) Application Layer:

Processes editing and synchronization requests.

iv) Cloud storage:

Stores document and related cites.

Working Process:

Web & mobile clients.

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HTTPS + RESTful Web APIs

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Cloud Backend Services / Application.

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Distributed Storage

Version Database.

Advantages:

* Supports real-time collaboration.
* Maintains document history

* Reduces the risk of data loss through backups.
* Provides security through access permissions.

CONCLUSION:

Real-time collaboration improves productivity, but network delays and simultaneous edits can create conflicts.

3) CASE 3: ONLINE FOOD ORDERING PLATFORM:

INTRODUCTION:

An online food ordering platform enables customers to browse menu and place food codes using website or mobile applications.

OBJECTIVES:

* Provide convenient online food ordering
* Store customer details, menu images and invoices,
* Process order efficiently through Web APIs.
* Protective customer information & transaction

M ARCHITECTURE:

Fronted Layer:

web browsers and mobile applications allow customers to place orders.

Restful API Layer:

Connects frontend clients with backend services.

Backend Layer:

Processes orders and manages customer information.

Cloud storage:

Stores menu images, invoices and related files.

Infrastructure Layer:

HTTPs, authentication tokens and role-based access control protect the platform.